Journal of Retailing and Consumer Services 79 (2024) 103867

Contents lists available at ScienceDirect

Journal of Retailing and Consumer Services

journal homepage: www.elsevier.com/locate/jretconser

Assessing the impact of AI-chatbot service quality on user e-brand loyalty through chatbot user trust, experience and electronic word of mouth

Muhammad Farrukh Shahzad [a,*], Shuo Xu [a], Xin An [b], Iqra Javed [c]

[a] College of Economics and Management, Beijing University of Technology, Beijing, 100124, PR China
[b] School of Economics and Management, Beijing Forestry University, Beijing, 100083, PR China
[c] Institute of Business & Management, University of Engineering and Technology, Lahore, 54000, Pakistan

ARTICLE INFO

Keywords:
AI-chatbot service quality
User e-brand loyalty
Chatbot user trust
Electronic word of mouth
Chatbot user experience

ABSTRACT

Based on the stimulus organism theory (S-O-R), the current study examines how the chatbot service quality influences e-brand loyalty among luxury fashion brand users and their adoption of it. In the current study, stimuli represent the chatbot service quality and chatbot awareness, prompting positive and negative emotional responses among Chinese users. However, these responses are influenced by the user's experience with chatbot usage, and trust is considered the organism in the S-O-R framework. In response to the user's ultimate trust, experience, electronic word of mouth (eWOM) and e-brand loyalty are enhanced. Similarly, our study framework is supported by the S-O-R theory. Data was gathered by convenience sampling and a cross-sectional research approach from 301 Chinese consumers of luxury fashion brands. The proposed relationships were tested using PLS-SEM through PLS-Smart software. The findings showed that AI-chatbot service quality significantly influences customer e-brand loyalty. Furthermore, chatbot user trust, experience, and electronic word of mouth significantly mediate the relationship between AI-chatbot service quality and customer e-brand loyalty. In a nutshell, applying S-O-R theory, our proposed model results provide valuable insights into the dynamics of AI-chatbot-driven interactions among Chinese luxury fashion brand consumers. Further, it explains the pivotal role of AI-chatbot service quality in fostering customer trust and e-brand loyalty. Moreover, the current study contributes to theoretical and practical knowledge.

1. Introduction

In today's digital era, the proliferation of e-commerce and online platforms has revolutionized how businesses interact with their luxury fashion brand consumers. The advent of artificial intelligence (AI) and chatbot technology has played a pivotal role in enhancing customer engagement and satisfaction for online luxury brands (Peltier et al., 2023). According to the (Mordor Intelligence reports, 2023), the Chatbot market is predicted to be worth USD 7.01 billion in 2024 and USD 20.81 billion by 2029, with a CAGR of 24.32% over the forecast period (2024–2029). AI-chatbots have become integral to customer service strategies, providing instantaneous responses, resolving queries, and guiding customers through their online journey. With the growing prevalence of AI-chatbots, it has become crucial for luxury brands to assess the impact of their service quality on customer e-brand loyalty (Chen et al., 2022). Currently, chatbots offer services in multiple sectors, such as sales function (41%), marketing (17%), and support (37%) (Misischia et al., 2022). These chatbots provide real-time assistance, streamline purchasing processes, and offer personalized recommendations while operating 24/7. Significantly, it increased revenues by an average of 67%, with chatbot engagements accounting for 26% of total sales (Kaushal and Yadav, 2023).

The AI technology behind these chatbots allows them to learn from user interactions, providing increasingly accurate and relevant responses over time (Kang and Choi, 2023). In particular, the adoption of AI-chatbots has surged in the rapidly growing Chinese e-commerce market, where companies seek to enhance customer experiences, improve service efficiency, and cultivate brand loyalty in the highly competitive online landscape (Kayeser Fatima et al., 2024). The Chinese luxury fashion industry has witnessed remarkable growth in recent years, driven by factors such as increasing internet penetration, widespread mobile usage, and a burgeoning middle class (Hollebeek et al., 2024). With many online platforms and products to choose from, customer loyalty has become a strategic imperative for businesses

* Corresponding author.
E-mail addresses: farrukhshahzad207@gmail.com (M.F. Shahzad), xushuo@bjut.edu.cn (S. Xu), anxin927@bjfu.edu.cn (X. An), iqra.javed@uet.edu.pk (I. Javed).

https://doi.org/10.1016/j.jretconser.2024.103867
Received 9 March 2024; Received in revised form 2 April 2024; Accepted 16 April 2024
Available online 22 April 2024

seeking to secure a sustainable competitive advantage. Several researchers (Bilgihan, 2016; S. Zhang et al., 2022) have provided insights into Chinese consumers' perceptions of fashion brands. This study chose China as its research setting for various compelling reasons. Akbari et al. (2022) discovered that Chinese luxury customers have stronger emotional attachments to brands, potentially leading to higher levels of positive brand association.

According to (Ittefaq et al., 2024), more than 60% of consumers are unlikely to purchase from the same brand again if they have an unpleasant experience, decreasing electronic word of mouth (eWOM) and electronic trust. Luxury fashion brands mostly use chatbots for their online services to customers to enhance quick access to the portal and stimulate them to build user trust due to positive experiences. Trust is critical for any online luxury fashion brand due to the mismanagement of consumers, who get annoyed and spread negative eWOM (Kilani and Rajaobelina, 2022). Moreover, e-brand loyalty is a critical factor for businesses operating in the online landscape. It reflects customers' emotional connection and commitment to a brand, influencing their repeat purchases, referrals, and overall lifetime value. The e-brand loyalty is closely associated with trust, satisfaction, and positive experiences with a brand, all of which can be influenced by AI-chatbot interactions (Kushwaha et al., 2021). When customers trust AI-chatbots to provide accurate information, resolve issues, and safeguard their data, it positively impacts their overall perception of the brand. Building and maintaining trust is vital for the long-term success of AI-chatbot implementations. As AI-technology continues to reshape how we interact with businesses and make purchasing decisions, e-loyalty has become a crucial metric for brands striving to succeed in the digital marketplace (C. Y. Li et al., 2023).

Electronic word of mouth, or the digital equivalent of word-of-mouth marketing, has gained immense significance in the age of social media and online communities (Yeo et al., 2022). Customers sharing their favorable experiences with AI-chatbots and e-brands can significantly influence potential customers' perceptions, further impacting e-brand loyalty. AI-chatbots can enhance the user experience by providing seamless navigation, personalized recommendations, and efficient issue resolution. A positive user experience increases customer retention rates and favorable eWOM (Kautish et al., 2023). Moreover, fewer studies have proven that chatbot service quality positively impacts user's e-brand loyalty (Ittefaq et al., 2024; S. Zhang et al., 2022). While, using S-O-R theory, fewer studies have explored the chatbot service quality and user's e-brand loyalty (Belhadi et al., 2023; Donthu et al., 2021). In the current study, S-O-R is used as a foundation theory to explore the external antecedent and consequences of the relationship between chatbot service quality and user's e-brand loyalty in the context of luxury fashion brands, and it also offers valuable practical and theoretical suggestions for the study.

Initially, limited researches (Ittefaq et al., 2024; Rahman et al., 2023; Y. Zhu et al., 2023) have investigated how chatbot service quality has affected the users' e-brand loyalty (see Table 1), but the current study has evaluated the chatbot service quality through the user's trust, experience, and eWOM perspective. Empirical researches (Chen, Lu, et al., 2023; Fan et al., 2023) into user willingness to reuse the chatbots and their e-brand loyalty is relevant. No prior research is available that tried to discuss the combined mediating effect of chatbot users' experience, eWOM, and user trust in the context of luxury fashion brand consumers. Hence, this study contributes to information technology, consumer behavior, digital marketing, and technology management literature. It gives marketers and consumers new insights into how to use chatbot technology for online services. The primary objective of this study is to comprehensively investigate the influence of AI-chatbot service quality on customer e-brand loyalty in China while considering the mediating roles of chatbot user trust, eWOM, and customer experience. Understanding these dynamics is crucial for the luxury brand to enhance its digital engagement strategies and cultivate lasting relationships with its online consumer base. This study aims to shed light on the mechanisms through which AI-chatbots affect e-brand loyalty, offering valuable insights for businesses looking to maximize the benefits of this technology in the Chinese market.

Table 1
Review of literature on chatbot service quality and e-brand loyalty through S-O-R theory.

Sr. No.	Author(s) & Year	Starting Point	Influencing Factors	Methodology	Model
1.	Mim et al. (2022)	Using the S-O-R model is better for analyzing customer responses to electronic word-of-mouth and brand loyalty.	Brand user trust, Brand attachment, Sustainable positioning.	Survey-393	S-O-R
2.	Rahman et al. (2023)	Using the S-O-R model is better for analyzing the customer experiences of luxury fashion brands.	Digital sensory cues, online platform, and online shopping experience.	Survey-1039	S-O-R
3.	Ittefaq et al. (2024)	Using the S-O-R model to analyze better corporate wrongdoings and betrayals among luxury fashion brand consumers.	Brand betrayal, avoidance, Reparation Brand, Inauthenticity, Rumination, Corporate wrongdoing.	Survey-215	S-O-R
4.	(Y. Zhu et al., 2023)	Using the S-O-R model is better for analyzing the role of chatbots in online travel agencies.	Travel chatbot, information quality, tourism internet marketing.	Survey-566	S-O-R
5.	Hashimoto and Karasawa (2018)	Analyzing the consumer power via the S-O-R model and its responses.	Consumer-related factors include behavioral response, i.e., WOM and willingness to buy.	Survey-385	S-O-R

2. Theoretical underpinning and hypotheses development

2.1. Stimulus-organism-response (S–O-R) theory

The S-O-R framework suggests that external stimuli lead to internal organismic responses, ultimately resulting in observable behavioral responses. Using the S-O-R theory, we develop a conceptual framework for assessing the impact of AI-chatbot service quality on customer e-brand loyalty through chatbot user trust, experience, and eWOM (X. Cheng et al., 2022). AI-chatbot service quality is the primary external stimulus (S). It encompasses various factors, such as the chatbot's responsiveness, accuracy, speed, and helpfulness. Chatbot users' trust is considered an internal organism (O), which is stated as an individual's emotional and mental condition referred to as their "organism." It can be influenced by perceived reliability, competence, transparency, and privacy in interactions with the chatbot (Fan et al., 2022). Chatbot user experience represents another internal response influenced by ease of use, personalization, and emotional satisfaction derived from chatbot interactions. The eWOM, or online word of mouth, is an intermediate internal response (R). It is driven by users' perceptions of chatbot service quality, trust, and their overall experience. Customer e-brand loyalty is the ultimate behavioral response in our study, indicating the extent to which customers are loyal to the brand online context due to their interactions with the AI-chatbot. It includes repeat purchases, positive reviews, recommendations, and brand advocacy (Pham et al., 2024).

Our study conceptual framework demonstrates how external stimuli (AI-chatbot service quality) influence internal organismic responses (chatbot user trust, chatbot user experience, and eWOM), which then impact the observable behavioral response (user e-brand loyalty). High AI-chatbot service quality (S) leads to increased chatbot user trust (O) because users perceive the chatbot as reliable, competent, and transparent (Rafiq et al., 2022). High AI-chatbot service quality (S) also contributes to a positive chatbot user experience (O) as users find the chatbot interactions easy, personalized, and emotionally satisfying. The positive chatbot user trust (O) and chatbot user experience (O) further stimulate the generation of positive electronic word of mouth (O) among users (Hernandez-Ortega and Ferreira, 2021). The positive electronic word of mouth (O) and the positive chatbot user trust (O) and chatbot user experience (O) collectively enhance higher levels of customer e-brand loyalty (R). Conversely, suppose the AI-chatbot service quality (S) is poor. In that case, it will lead to lower chatbot user trust (O), a negative chatbot user experience (O), and less favorable electronic word of mouth (O), ultimately resulting in reduced customer e-brand loyalty (R). Our study framework can guide and provide a theoretical foundation for investigating the relationship between AI-chatbot service quality and e-brand loyalty, as past studies provided evidence (Ho and Chow, 2023; Ittefaq et al., 2024).

2.2. AI-chatbot service quality and user e-brand loyalty

AI-chatbots are designed to simulate human conversation and provide information or assistance to users. Their role in customer service includes answering inquiries, resolving issues, and providing support around the clock (Y. Li et al., 2024). Service quality in the context of AI-chatbots encompasses various dimensions, including response accuracy, response time, user-friendliness, personalization, and overall effectiveness. Similarly, customer e-brand loyalty measures an attachment and commitment to a specific brand or company of individuals. It goes beyond mere satisfaction, reflecting a customer's willingness to continue buying from a brand and advocating for it (Yang et al., 2023). High-quality AI-chatbots enhance customer experiences by providing quick, accurate, and personalized responses. This positively impacts customer satisfaction, which, in turn, contributes to brand e-loyalty. Consistent and reliable AI-chatbot interactions build trust with customers. Trust is a critical factor in e-brand loyalty, as customers are likely to remain loyal to brands, they trust. Luxury fashion brands offer improved customer experiences, personalization, and consistency. Luxury fashion brands can foster stronger connections with their customers, ultimately leading to increased e-loyalty (Hsu and Lin, 2023).

AI-chatbots can consistently deliver a high level of service quality. They don't get tired, frustrated, or make mistakes due to fatigue, which can be true with human employees. This consistency ensures that customers receive the same level of service every time they interact online with the brand, reinforcing their trust and loyalty (Ruan and Mezei, 2022). Like consumers everywhere, luxury fashion consumers in China appreciate efficiency and convenience. AI-chatbots can provide quick responses and assist with tasks like order tracking, sizing recommendations, and returns processing. AI-chatbots can collect and analyze customer data to offer tailored product suggestions and content (M. Li and Wang, 2023). In the context of luxury fashion, where individual tastes and preferences matter greatly, this personalization can create a sense of exclusivity and make customers feel valued. This, in turn, can foster e-brand loyalty as customers feel that the brand understands and caters to their unique needs. AI-chatbots can gather valuable customer data and conversation insights (J. Y. S. Chang et al., 2023). This data can be used to refine marketing strategies, improve products, and identify emerging trends in the luxury fashion industry. Understanding customer preferences and behaviors can lead to more targeted marketing efforts that resonate with the audience, further strengthening e-brand loyalty (Niu and Mvondo, 2024). In light of these findings, the following hypothesis is presented.

H1. AI-chatbot service quality has a positive impact on user e-brand loyalty.

2.3. Mediating role of chatbot user trust

Trust is crucial in any customer-business relationship. Trust refers to the confidence and reliance customers place on the AI-chatbot to meet their needs and provide reliable information (Maroufkhani et al., 2022). Brand loyalty represents a customer's commitment and preference for a particular brand. It's the tendency to consistently choose one brand over others and even recommend it to others. High AI-chatbot service quality can positively influence customer trust because a chatbot that consistently provides accurate and helpful responses builds credibility (Chung et al., 2020). Customer trust, in turn, contributes to e-brand loyalty because customers are more likely to stick with a brand they trust, even if the interaction is with an AI system rather than a human. Luxury brands must actively build and maintain customer trust in their AI-chatbots (Nguyen et al., 2021). This can be achieved through transparency in AI usage, clear communication about the capabilities and limitations of chatbots, and consistently delivering on promises. When customers trust the chatbot and, by extension, the brand, they are more likely to remain loyal, make repeat purchases, and become brand advocates. Trust serves as a bridge that connects the quality of AI-chatbot services to customer loyalty. When customers trust the chatbot to provide accurate information and assistance, they are more likely to engage with the brand's chatbot, improving customer-brand interactions and loyalty (M. Li and Wang, 2023).

The e-brand loyalty is the extent to which customers repeatedly choose a particular brand over others. In the luxury fashion industry, brand loyalty is highly valued as it leads to repeat purchases and advocacy (Chen et al., 2022). Luxury brands need to maintain and nurture this loyalty among their customers. Chatbot trust implies that when customers perceive the AI-chatbot service as high in quality, it enhances their trust in the brand, subsequently leading to increased e-brand loyalty (Hsu and Lin, 2023). Consistency in AI-chatbot interactions, where the chatbot consistently provides high-quality assistance, contributes to trust-building. Customers feel secure in relying on the chatbot for their needs. Trust is also tied to customers' perceptions of the privacy and security of their data during interactions with the chatbot. Luxury fashion brands must ensure their AI-chatbots are secure and handle customer data carefully to maintain trust (Lee and Li, 2023). Understanding the role of trust in the Chinese luxury fashion market is crucial due to the unique cultural factors. In China, trust and reputation are highly valued in business relationships. Luxury fashion brands that trust Chinese consumers through AI-chatbot services will likely strengthen user e-loyalty (Chen, Lu, et al., 2023). Considering these concepts, the following hypothesis is developed for this study:

H2. Chatbot user trust positively mediates the relationship between AI-chatbot service quality and user e-brand loyalty.

2.4. Mediating role of electronic word of mouth

Electronic word of mouth is the process by which users share their experiences and opinions about a brand or product through online channels such as social media, review websites, and forums (Yeo et al., 2022). When users have positive experiences with AI-chatbot interactions, they are likelier to share their experiences through eWOM channels. This dissemination of positive information can reach a wider audience, potentially influencing others to try the brand's products or services (Belhadi et al., 2023). High-quality AI-chatbot interactions build trust and credibility in the brand. When customers read positive eWOM reviews from others, they are more likely to trust the brand and perceive it as reliable, which can strengthen their loyalty. Positive eWOM reviews can provide social validation to customers. When they see that others have had a good experience with the brand's chatbot, it reinforces their own positive perceptions and may encourage them to remain loyal (Kautish et al., 2023). This eWOM can also stimulate continuous engagement between the brand and its customers. Customers who are engaged with the brand through eWOM are more likely to stay connected, participate in loyalty programs, and make repeat purchases (Donthu et al., 2021).

A past study (Akbari et al., 2022) highlighted eWOM as a channel for sharing opinions, recommendations, and experiences about a brand or product online, often through social media, review platforms, or other digital channels. Positive eWOM indicates that customers talk favorably about the brand and its products. The eWOM is an intermediary variable that helps explain how or why AI-chatbot service quality influences e-brand loyalty (Vermeer et al., 2019). They can provide personalized recommendations, answer queries promptly, and create a sense of convenience. In the context of luxury fashion brands, where customer service and personalization are crucial, a positive AI-chatbot service experience can significantly impact customer satisfaction. When customers share positive feedback about their experiences on social media or review platforms, it can influence others to try this brand (Deryl et al., 2023). The Chinese market is unique and dynamic, particularly in luxury fashion. Consumers in China place a high value on brand image, status, and the luxury experience. Understanding this eWOM mediating relationship can inform marketing strategies for luxury fashion brands in China (Hwang et al., 2021). They may need to invest in AI-chatbot technologies and ensure that these services are of the highest quality. Furthermore, they can actively encourage and engage with customers on digital platforms to stimulate positive eWOM. Regular assessments and adjustments to the chatbot's performance may be necessary to maintain and improve customer satisfaction. This eWOM mediating relationship highlights the evolving landscape of customer engagement in the digital age and the strategic significance of AI-powered solutions in shaping brand perceptions and e-loyalty (Kudeshia & Kumar, 2017). Hence, from the above discussion, it is concluded that:

H3. Electronic word of mouth positively mediates the relationship between AI-chatbot service quality and user e-brand loyalty.

2.5. Mediating role of chatbot user experience

Chatbot user experience encompasses the overall interaction between a user and a product or service. In the context of AI-chatbots, user experience involves how smoothly and satisfyingly a customer can interact with the chatbot to meet their needs (Pizzi et al., 2021). When users have a positive experience with an AI-chatbot, they are likelier to perceive the service quality as high. If the chatbot consistently provides accurate and helpful responses, users will develop trust in the service. This perception of quality is a crucial driver of users' e-brand loyalty. A seamless and satisfying interaction with a chatbot contributes to overall user satisfaction (Van den Broeck et al., 2019). Satisfied customers are more likely to engage with the brand repeatedly and remain loyal. Positive user experiences create an emotional connection, making customers more attached to the brand. AI-chatbots may occasionally have limitations or errors (Rhim et al., 2022). However, a positive user experience can mitigate the impact of such failures. When users have a good experience with a chatbot, they are more forgiving of occasional mistakes, reducing the likelihood of negative brand sentiment. To ensure a positive user experience, businesses must prioritize user-centric design principles when developing and implementing chatbots (Skjuve et al., 2021).

AI-chatbots are increasingly used in China's luxury fashion industry to enhance customer service and engagement. User experience encompasses the overall interaction between customers and the AI-chatbot. It includes ease of use, convenience, personalization, and emotional satisfaction (Rapp et al., 2021). User e-brand loyalty in luxury fashion brands is characterized by customers' commitment to a specific brand, willingness to make repeat purchases and positive word-of-mouth recommendations. It is a crucial factor for luxury brands' long-term success and sustainability. A high-quality AI-chatbot service can engage customers effectively, providing personalized recommendations and an enjoyable shopping experience (Haugeland et al., 2022). A positive user experience during these interactions can lead to increased e-brand loyalty. When AI-chatbots consistently deliver accurate and relevant information, it can build trust and confidence in the brand. Customers with a positive user experience are more likely to trust the brand and develop a stronger emotional connection, which fosters loyalty (Borsci et al., 2022). User experience involves the convenience and efficiency of using AI-chatbots. If users find the AI-chatbot service easy to use and time-saving, it enhances their overall satisfaction. This satisfaction can translate into higher e-loyalty as customers are more likely to return to a brand that makes their shopping experience smoother (Nirala et al., 2022). A past study (Ait Baha et al., 2023) considered China's unique cultural and market dynamics when examining this relationship. Chinese luxury consumers value personalization, exclusivity, and brand prestige (Ittefaq et al., 2024). A seamless AI-chatbot interaction that caters to these preferences can significantly impact user experience and e-brand loyalty. From the above arguments, it is hypothesized that:

H4. Chatbot user experience positively mediates the relationship between AI-chatbot service quality and user e-brand loyalty.

Fig. 1 demonstrates the conceptual framework.

3. Research methodology

AI-based chatbot service quality has been considered a seamless integration of efficiency and personalization. In the context of Chinese luxury fashion brand consumers, the interaction between AI-powered chatbots and e-brand loyalty is analogous to creating a digital relationship. With their Mandarin (Chinese official language) skills and cultural insights, these intelligent conversation partners become trustworthy confidantes in the consumer's journey. AI-chatbot builds a story of dependability by providing personalized recommendations, answering questions, and changing in response to user feedback. As users enjoy the convenience and personalized interactions, a sense of e-brand loyalty develops. It is about developing a lasting attachment between consumers (users) and companies in the dynamic terrain of Chinese commerce, which is more than just about technology.

In this study, the sample population is the Chinese luxury fashion brand consumers who are well aware of the technology usage and its related benefits and are more committed to opting for it again. For this purpose, data were collected from two reliable platforms, Wenjuanxing and Josump, from November to December 2023 as an online questionnaire. The respondents are requested to complete the questionnaire by sharing their perspectives on the most frequently used AI-chatbot for online shopping from luxury fashion brands. Due to technological awareness and ease, Chinese consumers are more willing to use the online survey method than the conventional method (Ittefaq et al., 2024). The current study excludes luxury fashion brand consumers who use internet technology for online shopping but are unaware of using chatbot technology.

3.1. Measurement of items

In the current study, the research instrument and items were constructed based on existing research, and minor modifications were made to adjust them to the present context. The items were measured using a 7-point Likert scale and a rating scale ranging from strongly disagree =

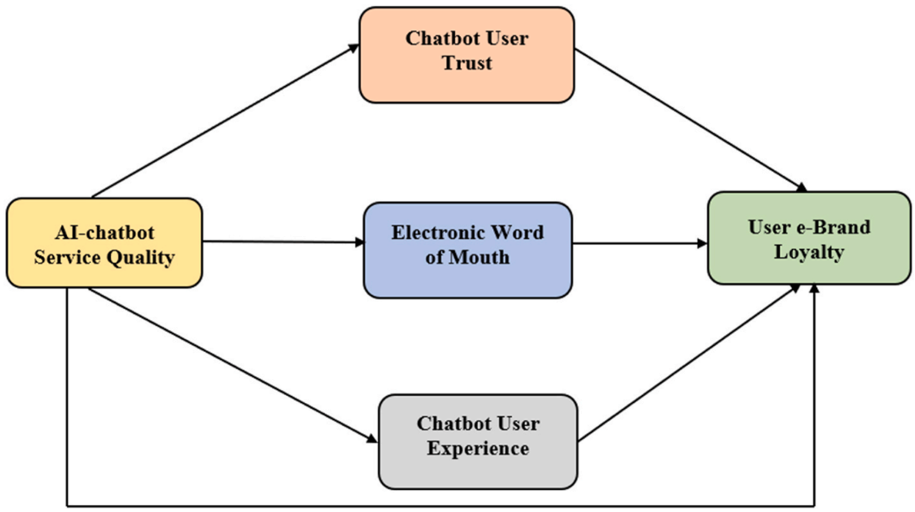

Fig. 1. Theoretical framework.

1, to strongly agree = 7. To measure the AI-chatbot service quality (AICSQ), eight items were adapted from (S. Zhang et al., 2022). The AICSQ is assessed using a 2-item scale of each dimension, such as tangibility, reliability, responsiveness, assurance, and empathy. We reinterpreted the tangibility dimension from equipment and physical facilities to the setting of the digital interface. For chatbot user trust (CUT), six items were taken from (Rajaobelina et al., 2021) scale measuring the user trust in terms of benevolence, perceived ability, and integrity. The chatbot user experience (CUE) scale is adapted from existing literature (X. Zhang et al., 2024) and slightly modified from retail business consumers to online shopping context. At the same time, the CUE was assessed with a 6-item scale from two dimensions: hedonic experiences and practical experiences. User e-brand loyalty (UeBL) was measured using a 4-item scale adapted from (Bilgihan, 2016). Finally, electronic word of mouth (eWOM) was evaluated using a 3-item scale adapted from (Kayeser Fatima et al., 2024) and slightly changed from the tourism sector to the e-shopping context for luxury brands. Moreover, we gathered feedback through a short pilot survey from 50 chatbot users, and based on their responses, we refined and completed the survey instrument. The reliability of all the items in the study was above 0.70, which was deemed satisfactory.

3.2. Data collection and sample characteristics

Through self-administered questionnaires, data is collected from two online platforms, Wenjuanxing and Josump. Wenjuanxing and Josump are extensively used platforms among researchers (Cao et al., 2021; W. Zhu et al., 2023), because of their high-quality data availability and results. Moreover, it allows scholars to gather data from multiple locations in China. The survey was conducted in November and December 2023. We divided the survey instrument into three sections. The first portion contained the purpose of the study. The second portion contained demographic information, including a question about respondents' frequency of using the chatbot for e-shopping from luxury fashion brands. The last portion compromised the variables under consideration. We collected data via convenience sampling and a cross-sectional survey. Convenience sampling is a non-probability sampling technique where researchers choose those individuals or subjects who are most easily accessible (Farrukh, Xu, Baheer, et al., 2023; Farrukh, Xu, Naveed, et al., 2023). Since it is difficult to contact each participant in the large population, a convenient sampling technique was used in this study as an efficient medium (Stratton, 2021). Participants in convenience sampling are selected according to their availability and desire to participate. Convenience sampling is widely used in online surveys since it's easy to reach respondents by email, social media, or other online methods. Although this technique has generalizability issues, it can be used excessively due to its ease of access to respondents and its relevancy to the measurement items (Ittefaq et al., 2024). So, the convenience sampling technique is not a great concern for the present study.

A screening question was used to identify those who shopped online but were not aware of chatbot technology. Respondents who answered a "No" were directed to complete the remaining questionnaire. A total of 420 questionnaires were circulated, and 384 were retrieved from consumers. However, all returned questionnaires were rigorously screened for missing values, multivariate outliers, and unengaged responses. Of the 83 cases that were rejected due to incomplete information, the remaining 301 were useable for analysis (78% response rate). In accordance with (Roscoe et al., 1975), a sample size of less than 500 and greater than 30 is highly recommended for marketing and behavioral science studies. Hence, we chose a sample size among 30–500 to ensure the validity of the present study. The details of the demographic factors are exhibited in Table 2.

4. Results and data analysis

The researchers used IBM SPSS 27.0 to assess CMB and demographics and Smart PLS 4 software to evaluate measurement and structural model techniques. Data was analyzed in two stages: (1) model fit, reliability, and validity, and (2) hypothesis testing.

4.1. Common method bias

To report CMB, the questionnaire outlined the study's strict anonymity policy. Respondents were instructed to be neutral and honest, with no correct or incorrect responses. We used Harman's single-factor technique, which provided a total variance of 14.01%, meeting the criteria. We performed a comprehensive collinearity test as per (Pereira et al., 2021) guideline. The results showed that all variance-inflated components were less than 3.3. Hence, CMB is not a matter of concern in the present research.

4.2. Measurement model evaluation

We assessed the constructs' reliability, discrimination, and convergent validity in the measurement model. While, Cronbach's alpha (α), composite reliability (CR), and average variance extracted (AVE) are the ways to assess the reliability of any construct. All the variables have

Table 2
Socio-demographic data (n = 301).

Demographics	Distribution	n = 301
Gender	Male	131 (44%)
	Female	170 (56%)
Age	18–24 years	90 (30%)
	25–34 years	130 (43%)
	35–44 years	45 (15%)
	44–54 years	25 (8%)
	Above 54	6 (2%)
Qualification	High School	80 (26%)
	Bachelor	133 (44%)
	Masters	89 (30%)
	PhD	0 (0%)
Income (RMB/Month)	<5000	45 (15%)
	5001–7000	50 (17%)
	7001–9000	50 (17%)
	9001–10,000	130 (43%)
	>10,000	26 (8%)
Frequency to use chatbots for e-shopping from luxury brands (in a year)	1 to 2 times	150 (50%)
	3 to 6	120 (40%)
	6 to 10	20 (6.6%)
	11 or above	11 (3.3%)
Brand-consumer relationship (years)	6 months or 1 year	151 (50%)
	1–2 years	90 (30%)
	3–4 years	51 (17%)
	4 and above	9 (3%)
Luxury fashion brands	Shanghai Tang	136 (45%)
	Chanal	90 (30%)
	Ochirly	45 (15%)
	Gucci	22 (7%)
	Burberry	8 (3%)

explained that values of Cronbach's Alpha (0.76–0.95) and composite reliability (0.86–0.96) are above the acceptable range of 0.70 (Sarstedt et al., 2014). Likewise, the values of AVE are also above the suggested acceptable range of 0.50, demonstrating the acceptable range for the reliability of the given construct. In PLS SEM, firstly, we accessed the convergent validity which is measured by factor loadings of each construct under study (Shahzad et al., 2023). Therefore, Table 3's results show that the factor loadings of each construct (0.66–0.85) are above the suggested threshold of 0.70, demonstrating satisfactory convergent validity (Hair et al., 2014). After the assessment of convergent validity in PLS-SEM, discriminant validity was measured, which states that it exists when the correlations between constructs are less than the square root of the AVE of the constructs as per (Bagozzi and Yi, 1989) criteria. Table 3 shows that all the constructs meet the criterion for good discriminant validity in the study.

Using the Fornell and Larcker criteria, discriminant validity was validated. Table 4 shows that the square roots of AVE exceeded its related correlations, indicating discriminant validity. The HTMT ratio, commonly used to evaluate discriminant validity (Fornell and Larcker, 1981) was applied in the current study. (Shahzad et al., 2024) advised a threshold of 0.90 for HTMT readings, although all results in Table 4 were lower. Therefore, discriminant validity was proven.

4.3. Structural model evaluation and hypotheses testing

After validating the data with the measurement model, the structural model analysis was used to test hypotheses. The bootstrapping resampling approach was used with a repeated sampling value of 5000 to determine the statistical significance of model coefficients (Martins et al., 2023). Table 5 shows empirical path coefficients, t-statistics, and hypothesis outcomes. (Hair et al., 2014) found that t-values of 1.96 or higher accurately estimate the relationship status between constructs. The exogenous variables' predictive importance was assessed using the endogenous latent variables' coefficients of determination (R2) (Shahzad et al., 2024).

After completing all prerequisites, structural correlations were evaluated using path coefficients (β), t-statistics (t), and significance (p). PLS-SEM results (Fig. 2 and Table 5) show a strong positive impact of AI-chatbot service quality on user e-brand loyalty ($\beta = 0.315$, $t = 7.662$, $p < 0.05$), supporting H1. Similarly, AI-chatbot service quality had directly and positively affected the chatbot user experience ($\beta = 0.389$, $t = 7.376$, $p < 0.05$), chatbot user trust ($\beta = 0.223$, $t = 4.251$, $p < 0.05$), and electronic word of mouth ($\beta = 0.704$, $t = 14.882$, $p < 0.05$). Moreover, user e-brand loyalty is directly and positively influenced by the chatbot user experience ($\beta = 0.232$, $t = 3.472$, $p < 0.05$), chatbot user trust ($\beta = 0.157$, $t = 2.917$, $p < 0.05$), and electronic word of mouth ($\beta = 0.270$, $t = 3.981$, $p < 0.05$). Hence, our all-direct results were statically validated and supported. Fig. 2, shows the statistical results of our study model.

4.4. Mediation analysis

The bootstrapping procedure in PLS-SEM demonstrated an indirect effect across all the constructs. SmartPLS software was used to assess the mediating effect of variables using bootstrapping and specific indirect effects. We examined the mediating role of chatbot user trust ($\beta = 0.035$, $t = 2.221$, $p < 0.05$), electronic word of mouth ($\beta = 0.190$, $t = 3.807$, $p < 0.05$), and chatbot user experience ($\beta = 0.090$, $t = 3.053$, $p < 0.05$) in between AICSQ and UeBL. Thus, all the mediating hypotheses H2, H3, and H4 were supported.

4.5. Model explanatory power

The model's explanatory power was assessed using (R^2), or coefficient of determination. We utilized the PLS technique to check R^2 and found that all scores were above the suggested limit of 0.10 (Bagozzi and Yi, 1989). Our study shows the R^2 values for chatbot user trust (0.050), chatbot user (eWOM) (0.496), chatbot user experience (0.151), and user e-brand loyalty (0.267). Moreover, the structural model evaluated goodness of fit using the SRMR criterion of 0.080. The SRMR score was below the limit of 0.067, indicating a well-fitted model that met the study's goals. We evaluated the impact of independent constructs on dependent constructs using f^2 values (Ittefaq et al., 2024). AI-chatbot service quality significantly influenced the chatbot user experience ($f^2 = 0.047$), chatbot user trust ($f^2 = 0.026$), and eWOM ($f^2 = 0.983$). Furthermore, chatbot user experience, chatbot user trust, and eWOM are affected by AI-chatbot service quality (($f^2 = 0.178$).

4.6. Model's predictive power

The model's predictive value was evaluated by cross-validated redundancy analysis (Q^2) (Hair et al., 2014; Ittefaq et al., 2024) used the blindfolding strategy to assess Q^2. (Shahzad et al., 2023) explains the model is predictive when the dependent variable's Q^2 value exceeds zero. Our study shows that the current model successfully demonstrates its predictive ability. As a result, we deemed all the values of Q^2 acceptable.

Table 3
Factor loadings.

Construct	Items		FL	α	CR	AVE	Source
AI-Chatbot Service Quality (AICSQ)				0.940	0.951	0.707	(S. Zhang et al., 2022)
	AICSQ1	The interface of chatbot is acceptable.	0.801				
	AICSQ2	The appearance of chatbot interface is appropriate.	0.914				
	AICSQ3	The chatbot fulfills the service promise.	0.900				
	AICSQ4	The chatbot provides the services within the specified time.	0.853				
	AICSQ5	The chatbot provides the services quickly.	0.857				
	AICSQ6	The chatbot has the intention to serve.	0.864				
	AICSQ7	The chatbot is trustworthy and supportive.	0.734				
	AICSQ8	The chatbot has the ability to provide services.	0.790				
User e-Brand Loyalty (UeBL)				0.831	0.887	0.664	Bilgihan (2016)
	UeBL1	As long as the current chatbots service is available, I will not move to the other platforms for shopping.	0.817				
	UeBL2	In my view, this platform is considered the best for online luxury brand shopping.	0.865				
	UeBL3	I feel this is my favorite online website.	0.780				
	UeBL4	I enjoy shopping through chatbots and recommend them to others.	0.795				
Chatbot User Experience (CUE)				0.939	0.952	0.767	(X. Zhang et al., 2024)
	CUE1	I have fun using chatbots to shop for luxury fashion brands.	0.876				
	CUE2	Using chatbots strengthens my desire to purchase.	0.885				
	CUE3	My friends and acquaintances use chatbots for luxury fashion brand shopping.	0.877				
	CUE4	I can easily buy what I need by using the chatbot platform.	0.873				
	CUE5	Luxury fashion brand services are convenient to use in many ways.	0.868				
	CUE6	I can save money and time by using chatbot for luxury fashion shopping.	0.875				
Chatbot User Trust (CUT)				0.953	0.966	0.850	Rajaobelina et al. (2021)
	CUT1	The chatbot is knowledgeable.	0.978				
	CUT2	The chatbot is reliable.	0.739				
	CUT3	The chatbot is competent and intelligent.	0.966				
	CUT4	The chatbot is pleasant.	0.949				
	CUT5	The chatbot is friendly.	0.957				
Electronic Word of Mouth (eWOM)				0.762	0.862	0.676	Kayeser Fatima et al. (2024)
	eWOM1	I have recommended the chatbot to a lot of people.	0.870				
	eWOM2	I talk up the chatbot to my friends.	0.829				
	eWOM3	I give a lot of advertising related to chatbot.	0.765				

Note: FL = Factor loadings, VIF = Variance inflation factor, CR = Composite reliability, AVE = Average variance extracted, α = Cronbach's alpha.

Table 4
Discriminant validity.

Constructs	AICSQ	CUE	UeBL	CUT	eWOM
Through Fornell–Larcker criterion					
AICSQ	**0.841**				
CUE	0.389	**0.876**			
UeBL	0.333	0.434	**0.815**		
CUT	0.223	0.476	0.338	**0.922**	
eWOM	0.704	0.475	0.421	0.264	**0.822**
Through Heterotrait-Monotrait ratio (HTMT)					
AICSQ					
CUE	0.413				
UeBL	0.377	0.491			
CUT	0.236	0.505	0.377		
eWOM	0.813	0.551	0.530	0.307	

Note: AICSQ = AI-chatbot Service Quality, UeBL = User e-Brand Loyalty, CUT = Chatbot User Trust, eWOM = Electronic Word of Mouth, CUE = Chatbot User Experience.

Table 5
Hypothesis testing.

Relationships	β-values	SD	t-values	p-values	Results
Direct effect					
AICSQ → UeBL	0.315	0.041	7.662	0.000	Supported
AICSQ → CUT	0.223	0.052	4.251	0.000	Supported
CUT → UeBL	0.157	0.054	2.917	0.004	Supported
AICSQ → eWOM	0.704	0.047	14.882	0.000	Supported
eWOM → UeBL	0.270	0.068	3.981	0.000	Supported
AICSQ → CUE	0.389	0.053	7.376	0.000	Supported
CUE → UeBL	0.232	0.065	3.572	0.000	
Mediation effect					
AICSQ → CUT → UeBL	0.035	0.016	2.221	0.027	Supported
AICSQ → eWOM → UeBL	0.190	0.050	3.807	0.000	Supported
AICSQ → CUE → UeBL	0.090	0.030	3.053	0.002	Supported

Note: AICSQ = AI-chatbot Service Quality, UeBL = User e-Brand Loyalty, CUT = Chatbot User Trust, eWOM = Electronic Word of Mouth, CUE = Chatbot User Experience, SD = Standard Deviation.

5. Discussion

This study verifies the application of S-O-R theory in the context of luxury e-shopping brands. Using the S-O-R model, this study confirms the predictors and results of chatbot user trust, user experience, and eWOM to promote e-brand loyalty in Chinese consumers. The study suggested that positive user experiences dramatically increase a user's electronic brand loyalty to chatbots. These positive encounters with chatbots are important in building trust and reliability between the user and the chatbot services. Customers' loyalty to the brand that offers these chatbot services strengthens as they interact positively with chatbots regarding query searches and information gathering. These results (Chen, Lu, et al., 2023) support our findings that chatbot user trust is influenced by chatbot service quality. Our findings are consistent with previous research by (Haugeland et al., 2022; Yeo et al., 2022), which found that chatbot user experience and eWOM can also be influenced by AI-self efficacy. After determining the stimuli of chatbot user trust, user experience, and eWOM, we investigated user reactions to chatbot service quality. Previous research has shown that Chinese consumers of luxury fashion brands exhibit a greater inclination towards these brands when they experience high service quality from chatbots

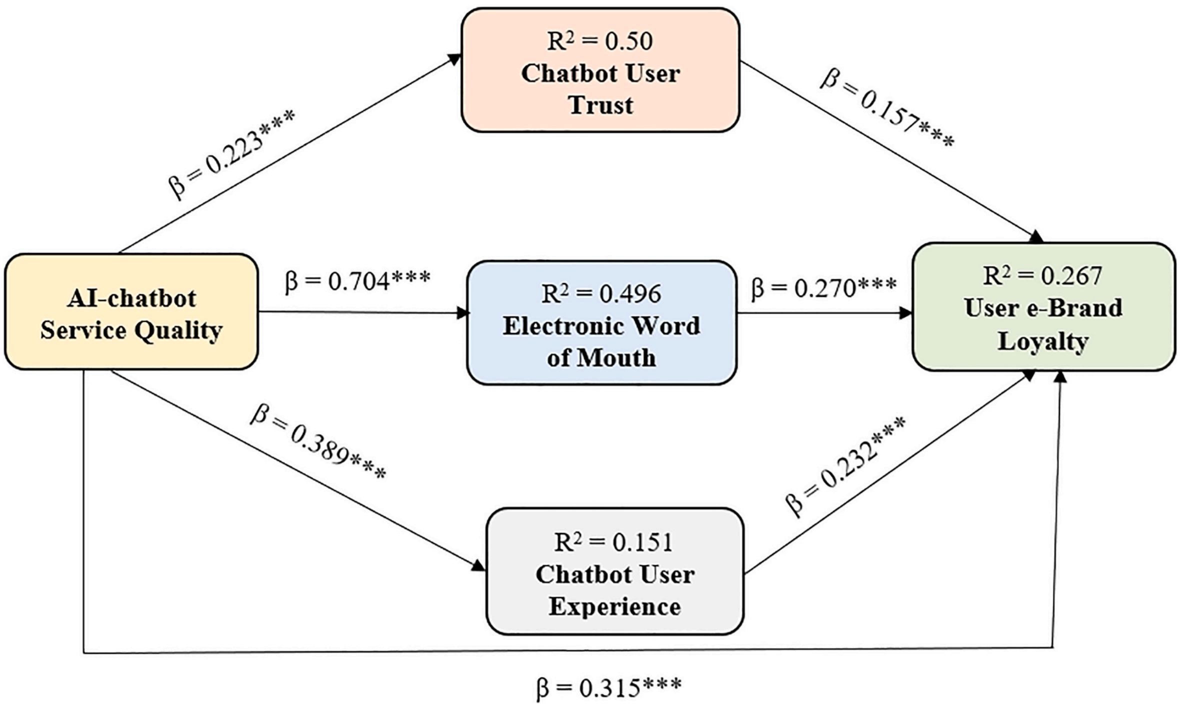

Fig. 2. PLS diagram (*p < 0.05; **p < 0.01; ***p < 0.001).

(Kautish et al., 2023).

Furthermore, the information technology literature has thoroughly researched the importance of AI-chatbot service quality, considering elements such as satisfaction, user trust, and performance (C. Y. Li et al., 2023). Previous studies (Elseidi and El-Baz, 2016; Hernandez-Ortega and Ferreira, 2021) have not addressed the mediation between chatbot user eWOM and user e-brand loyalty. Our findings support that user-friendly experiences and positive feedback from chatbot usage led to e-brand loyalty and subsequent recommendations for shopping portals (Huang et al., 2024). Similarly, the current study investigated the context of Chinese luxury e-shopping brands, specifically the mediating roles of CUE, CUT, and eWOM in the relationship between AI-chatbot service quality and user e-brand loyalty. By investigating these complex dynamics, the study assumes to identify critical factors that shape Chinese consumers' perceptions (Ittefaq et al., 2024) and loyalty outcomes in the context of luxury e-shopping brands, providing valuable insights into the specific elements driving user engagement (Hsu and Lin, 2023) and loyalty in this market segment.

5.1. *Theoretical implications*

The current study offers the theoretical implications from multiple sources of literature. First, this study introduces a unique viewpoint on how people perceive AI-chatbots and their impact on organizations. Scholars are exploring AI-chatbots' application in consumer behavior and marketing as they gain popularity. Prior research (Hsu and Lin, 2023) has focused on AI-chatbots as novel information systems but has missed their role as direct service providers to customers, particularly in the context of luxury e-shopping brands. Prior research (Nirala et al., 2022) has mostly examined AI-chatbots' influence as information systems on user attitudes and behaviors. However, considering AI-chatbots as service providers provides a unique perspective as the quality of their service can significantly affect customers' attitudes and organizational perceptions. Furthermore, this study fills a vacuum in empirical research evaluating the impact of AI-chatbot service quality on user e-brand loyalty. A thorough analysis of the existing AI-chatbot literature demonstrates that previous investigations used both theoretical and empirical methodologies. The empirical study on AI-chatbots focuses on consumer attitudes and behaviors, including adoption (Pentina et al., 2023), continued usage (Rafiq et al., 2022), emotional attachment (Niu and Mvondo, 2024), trust (Hsu and Lin, 2023), decision quality (Sarraf et al., 2023), and customer loyalty (Chen et al., 2022). However, research on AI-chatbots impact on consumers' e-brand loyalty is scarce (Bilgihan, 2016; S. Zhang et al., 2022). This study fills a gap by investigating the impact of AI-chatbot service quality on customer e-brand loyalty, adding to the research on AI-chatbot services.

Second, based on the S-O-R theory, this study provides nuanced implications for AI-chatbot service quality for Chinese luxury e-shopping brands, which has not been explored in prior studies (Hashimoto and Karasawa, 2018; Rahman et al., 2023). Applying the theoretical underpinning, AI-chatbot service quality is a stimulus, influencing chatbot user experience and trust (Organism). The study broadens S-O-R dimensions by examining how these internal states influence user e-brand loyalty and advocacy (Response). It emphasizes the importance of cultural context in determining the psychological effects of digital interactions, which contributes to S-O-R use in comprehending complex technology-consumer dynamics in the particular setting of luxury brands for e-shopping (Y. Zhu et al., 2023). The study indicates how technology interfaces functionally stimulate and profoundly influence cognitive and emotional responses, increasing customer loyalty in luxury markets. The current research contributes valuable insights to the S-O-R theory and information systems literature. Prior studies have primarily focused on these concepts in specific contexts such as hospitality (Kilani and Rajaobelina, 2022), the service sector (Chen, Yin, et al., 2023), and the education sector (Jo, 2023), but neglected the current study context. Hence, based on theoretical grounds, the S-O-R theory enhances the knowledge of luxury brands for e-shopping among Chinese consumers within the current background.

Third, the study highlights the chatbot interface quality by analyzing service quality dimensions such as reliability, empathy, tangibility, responsiveness, and assurance within the context of Chinese luxury brands for e-shoppers. While the Servqual model application is extensively used in the technology adoption intention (W. Chang and Park, 2024), actual use (Niu and Mvondo, 2024), and integration in multiple domains along with Chatbot user trust, experience, electronic word of mouth is under-researched.

Fourth, this study enhances the understanding by measuring the mediator role of chatbot user experience in the relationship between chatbot service quality and e-brand loyalty in a Chinese consumer context, supporting the arguments of prior scholars (Chen, Yin, et al., 2023; S. Zhang et al., 2022), enhancing the knowledge relevant to user e-brand loyalty perception outer the prior research context (Bilgihan, 2016). This research sheds light on users' positive reactions to chatbot experiences for e-brand loyalty. It confirms the role of eWOM in this situation, which is considered a novel insight into the current study. However, the results are supportive and add value to the knowledge of users' bipolar behavioral intentions for purchase (J. Y. S. Chang et al., 2023), and selective decision-making process (W. Chang and Park, 2024). Finally, this study contributes to the current literature by providing information on chatbot design and related services for e-brand consumers. This data can help researchers focus on these innovative topics for additional investigation in the literature.

5.2. Practical implications

Our study provides multiple novel managerial contributions to luxury e-brands that help them build strategies to maintain or enhance e-brand loyalty among Chinese chatbot users for e-shopping. Firstly, based on our results, we suggest that brand managers focus on service quality aspects such as assurance, reliability, relevance, and responsiveness to enhance the positive user experience, trust, and e-brand loyalty with these chatbots. Moreover, the tangibility factor can also be assessed in terms of chatbot design and features that help promote the quality of service due to a user-friendly interface. Hence, chatbot designers should pay more attention to this aspect (Kaushal and Yadav, 2023). This commitment to service quality directly influences user trust and satisfaction, ultimately leading to increased e-brand loyalty.

Secondly, while our findings highlight the importance of AI-chatbot service quality as a key determinant of user trust and e-brand loyalty towards Chatbot users. The strong mediating link of Chatbot user experience, electronic word of mouth, and user trust provide a compelling rationale for Chatbot developers and other companies that use this technology for e-businesses. It highlights the necessity of building chatbots with a positive and attractive aspect and creating a cool image for them. As chatbot adoption grows, organizations may struggle to sustain e-client loyalty. This leads to successfully developing unique chatbots with positive features and functions utilizing the Servqual framework, which can increase pleasure, user positive experience, trust, and e-brand loyalty (Zaato et al., 2023). Companies should focus on transparency, ensuring users understand the capabilities and limitations of chatbots. Clear communication about data privacy and security measures can alleviate user concerns and foster trust. Improving the user experience with AI-chatbots is paramount for fostering e-brand loyalty. Companies should implement strategies to incentivize users to share their positive experiences with AI-chatbots on social media platforms and review websites. This could include offering rewards or discounts for referrals, creating shareable content related to chatbot interactions, or actively engaging with users on social media to amplify positive feedback (Y. Cheng and Jiang, 2020).

Thirdly, electronic word of mouth significantly predicts e-brand loyalty for Chinese chatbot users. Luxury e-brand manufacturers should acknowledge the cultural and societal factors that influence their values, such as focusing on the native language rather than focusing on the English language in the chatbot design (Kayeser Fatima et al., 2024). This is the best way to attract consumers and build emotional loyalty. Fourthly, integrating high-quality AI-chatbot services into the brand experience provides a competitive advantage. Luxury brands that successfully combine new technology while knowing and respecting cultural distinctions will stand out in the competitive e-shopping landscape. This rapid change in technology and consumer preferences needs ongoing monitoring and adaptation. Managers should stay alert, gathering input on user experiences and adapting AI-chatbot strategies to meet changing consumer expectations and market trends (Castillo et al., 2020). Lastly, recognizing that the AI landscape is constantly evolving, organizations should prioritize continuous learning and adaptation in their chatbot strategies. This includes staying updated on emerging technologies, industry trends, and user preferences to remain competitive. Investing in employee training and development to keep pace with advancements in AI technology can help ensure that chatbot services remain relevant and effective in driving e-brand loyalty (Kang and Choi, 2023). By implementing these implications, organizations can harness AI-chatbots' potential to cultivate user trust, enhance experiences, stimulate positive eWOM, and foster long-term e-brand loyalty.

5.3. Conclusion

This study investigates how customers' views of AI-chatbot service quality affect users' e-brand loyalty. Using the S-O-R model that explains the service quality of AI-chatbots in terms of stimuli. It evaluates their impact on customer loyalty (response) through chatbot user trust, user experience, and user eWOM. The findings support the S-O-R model's applicability in the context of Chinese luxury e-shopping brands, contributing to research on chatbot service quality. This research study has a five-fold contribution; first, AI-chatbot service quality significantly influences customer e-brand loyalty. Second, chatbot user trust significantly mediates the relationship between AI-chatbot service quality and customer e-brand loyalty. Third, electronic word of mouth significantly mediates the relationship between AI-chatbot service quality and customer e-brand loyalty. Fourth, chatbot user experience significantly mediates the relationship between AI-chatbot service quality and customer e-brand loyalty. Furthermore, it broadens the model's understanding in this domain. The study offers significant insights for brand managers intending to improve user portal loyalty by suggesting increasing AI-chatbot service quality at various deployment levels.

5.4. Limitations and future directions

This study highlights certain shortcomings and suggests areas for additional research. First, we found that Chinese customers of different ages and regions have varying opinions towards luxury e-shopping brands and share different experiences and feedback related to chatbots. Future research could examine how different age groups react to eWOM and share their portal experiences. Second, the study followed a cross-sectional design. To establish causal linkages, longitudinal data collection would be exciting, allowing for investigating changes in users' attitudes and behaviors over time. Future studies could investigate the change in customers' pre- and post-experiences with purchases based on eWOM and its related trust in an AI-chatbot. Third, in this study, we collected data via convenience sampling. Participants in convenience sampling are selected according to their availability and desire to participate. Future researchers would use other probability sampling methods, such as simple random sampling or stratified sampling, to generalize the findings. Fourth, we solely emphasized e-brand loyalty as an outcome variable in the model. Future researchers can investigate the further role of e-brand loyalty in bipolar behavioral intentions (revisiting or switching the chatbot); the ultimate reason is to expand the new insights of the topic. Fifth, the current investigation has only discussed the mediating role of CUE, CUT, and eWOM. Still, future researchers can examine the model with multiple moderators, such as users' ethical perceptions and ethical beliefs towards chatbot usage, which will enhance the authenticity of the current research. Finally, the study's focus on China may limit its applicability to other contexts. Scholars can apply this approach to analyze cross-cultural or comparative contexts.

CRediT authorship contribution statement

Muhammad Farrukh Shahzad: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Shuo Xu:** Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition, Conceptualization. **Xin An:** Data curation, Investigation, Methodology, Resources, Supervision, Visualization. **Iqra Javed:** Writing – review & editing, Visualization, Resources, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This work received financial support from the National Natural Science Foundation of China under grant number 72074014.

References

Ait Baha, T., El Hajji, M., Es-Saady, Y., Fadili, H., 2023. The impact of educational chatbot on student learning experience. Educ. Inf. Technol., 0123456789 https://doi.org/10.1007/s10639-023-12166-w.

Akbari, M., Foroudi, P., Zaman Fashami, R., Mahavarpour, N., Khodayari, M., 2022. Let us talk about something: the evolution of e-WOM from the past to the future. J. Bus. Res. 149 (December 2021), 663–689. https://doi.org/10.1016/j.jbusres.2022.05.061.

Bagozzi, R.P., Yi, Y., 1989. The degree of intention formation as a moderator of the attitude-behavior relationship. Soc. Psychol. Q. 266–279.

Belhadi, A., Kamble, S., Benkhati, I., Gupta, S., Mangla, S.K., 2023. Does strategic management of digital technologies influence electronic word-of-mouth (eWOM) and customer loyalty? Empirical insights from B2B platform economy. J. Bus. Res. 156 (December 2022), 113548 https://doi.org/10.1016/j.jbusres.2022.113548.

Bilgihan, A., 2016. Gen y customer loyalty in online shopping: an integrated model of trust, user experience and branding. Comput. Hum. Behav. 61, 103–113. https://doi.org/10.1016/j.chb.2016.03.014.

Borsci, S., Malizia, A., Schmettow, M., van der Velde, F., Tariverdiyeva, G., Balaji, D., Chamberlain, A., 2022. The chatbot usability scale: the design and pilot of a usability scale for interaction with AI-based conversational agents. Personal Ubiquitous Comput. 26 (1), 95–119. https://doi.org/10.1007/s00779-021-01582-9.

Cao, J., Yang, T., Lai, I.K.W., Wu, J., 2021. Is online education more welcomed during COVID-19? An empirical study of social impact theory on online tutoring platforms. Int. J. Electr. Eng. Educ. https://doi.org/10.1177/0020720920984001.

Castillo, D., Canhoto, A.I., Said, E., Castillo, D., Canhoto, A.I., 2020. The dark side of AI-powered service interactions : exploring the process of co-destruction from the customer perspective. Serv. Ind. J. 0 (0), 1–26. https://doi.org/10.1080/02642069.2020.1787993.

Chang, J.Y.S., Cheah, J.H., Lim, X.J., Morrison, A.M., 2023. One pie, many recipes: the role of artificial intelligence chatbots in influencing Malaysian solo traveler purchase intentions. Tourism Manag. Perspect. 49 (April 2022), 101191 https://doi.org/10.1016/j.tmp.2023.101191.

Chang, W., Park, J., 2024. A comparative study on the effect of ChatGPT recommendation and AI recommender systems on the formation of a consideration set. J. Retailing Consum. Serv. 78 (February), 103743 https://doi.org/10.1016/j.jretconser.2024.103743.

Chen, Q., Gong, Y., Lu, Y., Tang, J., 2022. Classifying and measuring the service quality of AI chatbot in frontline service. J. Bus. Res. 145 (July 2021), 552–568. https://doi.org/10.1016/j.jbusres.2022.02.088.

Chen, Q., Lu, Y., Gong, Y., Xiong, J., 2023a. Can AI chatbots help retain customers? Impact of AI service quality on customer loyalty. https://doi.org/10.1108/INTR-09-2021-0686. In Internet Research (Vol. 33, Issue 6).

Chen, Q., Yin, C., Gong, Y., 2023b. Would an AI chatbot persuade you: an empirical answer from the elaboration likelihood model. Information Technology and People 72001085. https://doi.org/10.1108/ITP-10-2021-0764.

Cheng, X., Zhang, X., Cohen, J., Mou, J., 2022. Human vs. AI: understanding the impact of anthropomorphism on consumer response to chatbots from the perspective of trust and relationship norms. Inf. Process. Manag. 59 (3), 102940 https://doi.org/10.1016/j.ipm.2022.102940.

Cheng, Y., Jiang, H., 2020. How do AI-driven chatbots impact user experience ? Examining gratifications , perceived privacy risk , satisfaction , loyalty , and continued use. J. Broadcast. Electron. Media 00 (00), 1–23. https://doi.org/10.1080/08838151.2020.1834296.

Chung, M., Ko, E., Joung, H., Kim, S.J., 2020. Chatbot e-service and customer satisfaction regarding luxury brands. J. Bus. Res. 117 (November 2017), 587–595. https://doi.org/10.1016/j.jbusres.2018.10.004.

Deryl, M. Ds, Verma, S., Srivastava, V., 2023. How does AI drive branding? Towards an integrated theoretical framework for AI-driven branding. Int. J. Inf. Manag. Data Insights 3 (2), 100205. https://doi.org/10.1016/j.jjimei.2023.100205.

Donthu, N., Kumar, S., Pandey, N., Pandey, N., Mishra, A., 2021. Mapping the electronic word-of-mouth (eWOM) research: a systematic review and bibliometric analysis. J. Bus. Res. 135 (February), 758–773. https://doi.org/10.1016/j.jbusres.2021.07.015.

Elseidi, R.I., El-Baz, D., 2016. Electronic word of mouth effects on consumers' brand attitudes, brand image. In: International Conference on Restructuring of the Global Economy (ROGE), vol. 7. University of, Oxford, UK, pp. 268–276, 5.

Fan, H., Gao, W., Han, B., 2023. Are AI chatbots a cure-all? The relative effectiveness of chatbot ambidexterity in crafting hedonic and cognitive smart experiences. J. Bus. Res. 156 (November 2022), 113526 https://doi.org/10.1016/j.jbusres.2022.113526.

Fan, H., Han, B., Gao, W., 2022. (Im)Balanced customer-oriented behaviors and AI chatbots' Efficiency–Flexibility performance: the moderating role of customers' rational choices. J. Retailing Consum. Serv. 66, 102937 https://doi.org/10.1016/j.jretconser.2022.102937 (August 2021).

Farrukh, M., Xu, S., Baheer, R., Ahmad, W., 2023. Unveiling the role of supply chain parameters approved by blockchain technology towards firm performance through trust : The moderating role of government support. Heliyon 9 (11), e21831. https://doi.org/10.1016/j.heliyon.2023.e21831.

Farrukh, M., Xu, S., Naveed, W., Nusrat, S., 2023. Investigating the impact of artificial intelligence on human resource functions in the health sector of China : A mediated moderation model. Heliyon 9 (11), e21818. https://doi.org/10.1016/j.heliyon.2023.e21818.

Fornell, C., Larcker, D.F., 1981. Evaluating structural equation models with unobservable variables and measurement error. J. Market. Res. 18 (1), 39. https://doi.org/10.2307/3151312.

Hair, J.F., Sarstedt, M., Hopkins, L., Kuppelwieser, V.G., 2014. Partial least squares structural equation modeling (PLS-SEM): an emerging tool in business research. Eur. Bus. Rev. 26 (2), 106–121. https://doi.org/10.1108/EBR-10-2013-0128.

Hashimoto, T., Karasawa, K., 2018. Impact of consumer power on consumers' reactions to corporate transgression. PLoS One 13 (5), 1–14. https://doi.org/10.1371/journal.pone.0196819.

Haugeland, I.K.F., Følstad, A., Taylor, C., Alexander, C., 2022. Understanding the user experience of customer service chatbots: an experimental study of chatbot interaction design. Int. J. Hum. Comput. Stud. 161. https://doi.org/10.1016/j.ijhcs.2022.102788. April 2021).

Hernandez-Ortega, B., Ferreira, I., 2021. How smart experiences build service loyalty: the importance of consumer love for smart voice assistants. Psychol. Market. 38 (7), 1122–1139. https://doi.org/10.1002/mar.21497.

Ho, S.P.S., Chow, M.Y.C., 2023. The role of artificial intelligence in consumers' brand preference for retail banks in Hong Kong. J. Financ. Serv. Market., 0123456789 https://doi.org/10.1057/s41264-022-00207-3.

Hollebeek, L.D., Menidjel, C., Sarstedt, M., Jansson, J., Urbonavicius, S., 2024. Engaging consumers through artificially intelligent technologies: systematic review, conceptual model, and further research. Psychol. Market. 1–19. https://doi.org/10.1002/mar.21957. December 2023.

Hsu, C.L., Lin, J.C.C., 2023. Understanding the user satisfaction and loyalty of customer service chatbots. J. Retailing Consum. Serv. 71 (October 2022), 103211 https://doi.org/10.1016/j.jretconser.2022.103211.

Huang, D., Markovitch, D.G., Stough, R.A., 2024. Can chatbot customer service match human service agents on customer satisfaction? An investigation in the role of trust. J. Retailing Consum. Serv. 76 (October 2023), 103600 https://doi.org/10.1016/j.jretconser.2023.103600.

Hwang, J., Choe, J.Y., Jacey, Kim, H.M., Kim, J.J., 2021. Human baristas and robot baristas: how does brand experience affect brand satisfaction, brand attitude, brand attachment, and brand loyalty? Int. J. Hospit. Manag. 99 (August), 103050 https://doi.org/10.1016/j.ijhm.2021.103050.

Ittefaq, H., Akhtar, N., Siddiqi, U.I., Islam, T., Kuzior, A., 2024. The betrayal puzzle: unraveling the connection between inauthenticity, corporate wrongdoing and brand betrayal with avoidance and reparation. J. Retailing Consum. Serv. 76 (October 2023), 103597 https://doi.org/10.1016/j.jretconser.2023.103597.

Jo, H., 2023. Understanding AI tool engagement: a study of ChatGPT usage and word-of-mouth among university students and office workers. Telematics Inf. 85 (June), 102067 https://doi.org/10.1016/j.tele.2023.102067.

Kang, J.Y.M., Choi, D., 2023. Artificial intelligence-powered digital solutions in the fashion industry: a mixed-methods study on AI-based customer services. Int. J. Fash. Des. Technol. Educ. 1–15. https://doi.org/10.1080/17543266.2023.2261019.

Kaushal, V., Yadav, R., 2023. Learning successful implementation of Chatbots in businesses from B2B customer experience perspective. Concurrency Comput. Pract. Ex. 35 (1), 1–18. https://doi.org/10.1002/cpe.7450.

Kautish, P., Purohit, S., Filieri, R., Dwivedi, Y.K., 2023. Examining the role of consumer motivations to use voice assistants for fashion shopping: the mediating role of awe experience and eWOM. Technol. Forecast. Soc. Change 190, 122407. https://doi.org/10.1016/j.techfore.2023.122407. August 2022.

Kayeser Fatima, J., Khan, M.I., Bahmannia, S., Chatrath, S.K., Dale, N.F., Johns, R., 2024. Rapport with a chatbot? The underlying role of anthropomorphism in socio-

cognitive perceptions of rapport and e-word of mouth. J. Retailing Consum. Serv. 77 (January) https://doi.org/10.1016/j.jretconser.2023.103666.
Kilani, N., Rajaobelina, L., 2022. Impact of live chat service quality on behavioral intentions and relationship quality: a meta-analysis. Int. J. Hum. Comput. Interact. 0 (0), 1–28. https://doi.org/10.1080/10447318.2022.2144126.
Kudeshia, C., Kumar, A., 2017. Social eWOM: does it affect the brand attitude and purchase intention of brands? Manag. Res. Rev. 40 (3), 310–330. https://doi.org/10.1108/MRR-07-2015-0161.
Kushwaha, A.K., Kumar, P., Kar, A.K., 2021. What impacts customer experience for B2B enterprises on using AI-enabled chatbots? Insights from Big data analytics. Ind. Market. Manag. 98 (September), 207–221. https://doi.org/10.1016/j.indmarman.2021.08.011.
Lee, K.W., Li, C.Y., 2023. It is not merely a chat: transforming chatbot affordances into dual identification and loyalty. J. Retailing Consum. Serv. 74 (129), 103447 https://doi.org/10.1016/j.jretconser.2023.103447.
Li, C.Y., Fang, Y.H., Chiang, Y.H., 2023. Can AI chatbots help retain customers? An integrative perspective using affordance theory and service-domain logic. Technol. Forecast. Soc. Change 197 (151), 122921. https://doi.org/10.1016/j.techfore.2023.122921.
Li, M., Wang, R., 2023. Chatbots in e-commerce: the effect of chatbot language style on customers' continuance usage intention and attitude toward brand. J. Retailing Consum. Serv. 71 (August 2022) https://doi.org/10.1016/j.jretconser.2022.103209.
Li, Y., Li, Y., Chen, Q., Chang, Y., 2024. Humans as teammates: the signal of human–AI teaming enhances consumer acceptance of chatbots. Int. J. Inf. Manag. 76 (February), 102771 https://doi.org/10.1016/j.ijinfomgt.2024.102771.
Maroufkhani, P., Asadi, S., Ghobakhloo, M., Jannesari, M.T., Ismail, W.K.W., 2022. How do interactive voice assistants build brands' loyalty? Technol. Forecast. Soc. Change 183 (December 2021), 121870. https://doi.org/10.1016/j.techfore.2022.121870.
Martins, J.M., Shahzad, M.F., Javed, I., 2023. Assessing the impact of workplace harassment on turnover intention: evidence from the banking industry. Emerg. Sci. J. 7 (5), 1699–1722. https://doi.org/10.28991/ESJ-2023-07-05-016.
Mim, K.B., Jai, T., Lee, S.H., 2022. The influence of sustainable positioning on eWOM and brand loyalty: analysis of credible sources and transparency practices based on the S-O-R model. Sustainability 14 (19). https://doi.org/10.3390/su141912461.
Misischia, C.V., Poecze, F., Strauss, C., 2022. Chatbots in customer service: their relevance and impact on service quality. Procedia Comput. Sci. 201 (C), 421–428. https://doi.org/10.1016/j.procs.2022.03.055.
Mordor Intelligence reports, 2023. Chatbot Market Size & Share Analysis - Growth Trends & Forecasts. *2024 - 2029)*.
Nguyen, D.M., Chiu, Y.T.H., Le, H.D., 2021. Determinants of continuance intention towards banks' chatbot services in vietnam: a necessity for sustainable development. Sustainability 13 (14), 1–24. https://doi.org/10.3390/su13147625.
Nirala, K.K., Singh, N.K., Purani, V.S., 2022. A survey on providing customer and public administration based services using AI: chatbot. In: Multimedia Tools and Applications, vol. 81. Springer, US. https://doi.org/10.1007/s11042-021-11458-y. Issue 16.
Niu, B., Mvondo, G.F.N., 2024. I Am ChatGPT, the ultimate AI Chatbot! Investigating the determinants of users' loyalty and ethical usage concerns of ChatGPT. J. Retailing Consum. Serv. 76 (May 2023), 103562 https://doi.org/10.1016/j.jretconser.2023.103562.
Peltier, J.W., Dahl, A.J., Schibrowsky, J.A., 2023. Artificial intelligence in interactive marketing: a conceptual framework and research agenda. J. Res. Indian Med. https://doi.org/10.1108/JRIM-01-2023-0030.
Pentina, I., Xie, T., Hancock, T., Bailey, A., 2023. Consumer–machine relationships in the age of artificial intelligence: systematic literature review and research directions. Psychol. Market. 40 (8), 1593–1614. https://doi.org/10.1002/mar.21853.
Pereira, T., Limberger, D.P.F., Ardigó, D.C.M., 2021. The moderating effect of the need for interaction with a service employee on purchase intention in chatbots. Telemat. Inform. Rep. (December 2021), 1–4. https://doi.org/10.1016/j.teler.2022.100003.
Pham, H.C., Duong, C.D., Nguyen, G.K.H., 2024. What drives tourists' continuance intention to use ChatGPT for travel services? A stimulus-organism-response perspective. J. Retailing Consum. Serv. 78 (February), 103758 https://doi.org/10.1016/j.jretconser.2024.103758.
Pizzi, G., Scarpi, D., Pantano, E., 2021. Artificial intelligence and the new forms of interaction: who has the control when interacting with a chatbot? J. Bus. Res. 129 (November 2020), 878–890. https://doi.org/10.1016/j.jbusres.2020.11.006.
Rafiq, F., Dogra, N., Adil, M., Wu, J.Z., 2022. Examining consumer's intention to adopt AI-chatbots in tourism using partial least squares structural equation modeling method. Mathematics 10 (13). https://doi.org/10.3390/math10132190.
Rahman, M.S., Bag, S., Hossain, M.A., Abdel Fattah, F.A.M., Gani, M.O., Rana, N.P., 2023. The new wave of AI-powered luxury brands online shopping experience: the role of digital multisensory cues and customers' engagement. J. Retailing Consum. Serv. 72 (January), 103273 https://doi.org/10.1016/j.jretconser.2023.103273.
Rajaobelina, L., Prom Tep, S., Arcand, M., Ricard, L., 2021. Creepiness: its antecedents and impact on loyalty when interacting with a chatbot. Psychol. Market. 38 (12), 2339–2356. https://doi.org/10.1002/mar.21548.
Rapp, A., Curti, L., Boldi, A., 2021. The human side of human-chatbot interaction: a systematic literature review of ten years of research on text-based chatbots. Int. J. Hum. Comput. Stud. 151 (March), 102630 https://doi.org/10.1016/j.ijhcs.2021.102630.
Rhim, J., Kwak, M., Gong, Y., Gweon, G., 2022. Application of humanization to survey chatbots: change in chatbot perception, interaction experience, and survey data quality. Comput. Hum. Behav. 126 (April 2021), 107034 https://doi.org/10.1016/j.chb.2021.107034.
Roscoe, A.M., Lang, D., Sheth, J.N., 1975. Follow-up methods, questionnaire length, and market differences in mail surveys. J. Market. 39 (2), 20. https://doi.org/10.2307/1250111.
Ruan, Y., Mezei, J., 2022. When do AI chatbots lead to higher customer satisfaction than human frontline employees in online shopping assistance? Considering product attribute type. J. Retailing Consum. Serv. 68 (March) https://doi.org/10.1016/j.jretconser.2022.103059.
Sarraf, S., Kar, A.K., Janssen, M., 2023. How do system and user characteristics, along with anthropomorphism, impact cognitive absorption of chatbots – introducing SUCCAST through a mixed methods study. Decis. Support Syst. 178 (November 2023), 114132 https://doi.org/10.1016/j.dss.2023.114132.
Sarstedt, M., Ringle, C.M., Hair, J.F., 2014. PLS-SEM: looking back and moving forward. Long. Range Plan. 47 (3), 132–137. https://doi.org/10.1016/j.lrp.2014.02.008.
Shahzad, M.F., Xu, S., Baheer, R., 2024. Assessing the Factors Influencing the Intention to Use Information and Communication Technology Implementation and Acceptance in China's Education Sector, pp. 1–15. https://doi.org/10.1057/s41599-024-02777-0.
Shahzad, M.F., Xu, S., Lim, W.M., Hasnain, M.F., Nusrat, S., 2024. Cryptocurrency awareness, acceptance, and adoption: the role of trust as a cornerstone. Humanit. Soc. Sci. Commun. https://doi.org/10.1057/s41599-023-02528-7.
Shahzad, M.F., Xu, S., Rehman, O., Javed, I., 2023. Impact of gamification on green consumption behavior integrating technological awareness , motivation , enjoyment and virtual CSR. Sci. Rep. 1–18. https://doi.org/10.1038/s41598-023-48835-6.
Skjuve, M., Følstad, A., Fostervold, K.I., Brandtzaeg, P.B., 2021. My chatbot companion - a study of human-chatbot relationships. Int. J. Hum. Comput. Stud. 149 (January) https://doi.org/10.1016/j.ijhcs.2021.102601.
Stratton, S.J., 2021. Population research: convenience sampling strategies. Prehospital Disaster Med. 36 (4), 373–374. https://doi.org/10.1017/S1049023X21000649.
Van den Broeck, E., Zarouali, B., Poels, K., 2019. Chatbot advertising effectiveness: when does the message get through? Comput. Hum. Behav. 98 (April), 150–157. https://doi.org/10.1016/j.chb.2019.04.009.
Vermeer, S.A.M., Araujo, T., Bernritter, S.F., van Noort, G., 2019. Seeing the wood for the trees: how machine learning can help firms in identifying relevant electronic word-of-mouth in social media. Int. J. Res. Market. 36 (3), 492–508. https://doi.org/10.1016/j.ijresmar.2019.01.010.
Yang, B., Sun, Y., Shen, X.L., 2023. Understanding AI-based customer service resistance: a perspective of defective AI features and tri-dimensional distrusting beliefs. Inf. Process. Manag. 60 (3), 103257 https://doi.org/10.1016/j.ipm.2022.103257.
Yeo, S.F., Tan, C.L., Kumar, A., Tan, K.H., Wong, J.K., 2022. Investigating the impact of AI-powered technologies on Instagrammers' purchase decisions in digitalization era–A study of the fashion and apparel industry. Technol. Forecast. Soc. Change 177 (October 2021), 121551. https://doi.org/10.1016/j.techfore.2022.121551.
Zaato, S.G., Zainol, N.R., Khan, S., Rehman, A.U., 2023. The Mediating Role of Customer Satisfaction between Antecedent Factors and Brand Loyalty for the Shopee Application, vols. 1–22. https://doi.org/10.3390/bs13070563.
Zhang, S., Hu, Z., Li, X., Ren, A., 2022. The impact of service principal (service robot vs. human staff) on service quality: the mediating role of service principal attribute. J. Hospit. Tourism Manag. 52 (January), 170–183. https://doi.org/10.1016/j.jhtm.2022.06.014.
Zhang, X., Park, Y., Park, J., Zhang, H., 2024. Demonstrating the influencing factors and outcomes of customer experience in omnichannel retail. J. Retailing Consum. Serv. 77 (September 2023), 103622 https://doi.org/10.1016/j.jretconser.2023.103622.
Zhu, W., Lu, S., Huang, P., Hu, X., Wei, J., 2023. Predictors of coping behavior during the COVID-19 pandemic: evidence from China. J. Contingencies Crisis Manag. 31 (4), 797–808. https://doi.org/10.1111/1468-5973.12481.
Zhu, Y., Zhang, R., Renata, Zou, Y., Jin, D., 2023. Investigating customers' responses to artificial intelligence chatbots in online travel agencies: the moderating role of product familiarity. J. Hosp. Tour. Technol. 14 (2), 208–224. https://doi.org/10.1108/JHTT-02-2022-0041.